

Algorithms Problem Solving Level 2

Multiplication Table 1 To 10

The background of the slide is a photograph of a modern building's interior. It features a large, curved glass wall that reflects the sky and surrounding environment. In the foreground, the silhouettes of several people are visible, some standing and some sitting, looking towards the glass wall. The overall lighting is bright, suggesting a sunny day.

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Multiplication Table 1 To 10

Definition

- ◆ This program prints a multiplication table containing the numbers from 1 to 10.
- ◆ It uses nested `for` loops to generate the rows and columns of the table.
- ◆ Each cell is calculated by multiplying the current row number by the current column number.

Explanation

- ◆ ``PrintMultiplicationTableHeader()`` prints the title and the numbers from 1 to 10 at the top of the table.
- ◆ The first ``for`` loop prints the column numbers:
``1 2 3 ... 10``
- ◆ ``PrintMultiplicationTable()`` prints the complete table.
- ◆ The outer ``for`` loop controls the rows:
``i = 1`` to ``10``
- ◆ The inner ``for`` loop controls the columns:
``j = 1`` to ``10``
- ◆ The expression ``i * j`` calculates the value for each cell.
- ◆ ``\t`` is used to create horizontal spacing between values.
- ◆ The ``|`` character separates the row number from the multiplication results.

Steps in main()

- 1 Print the table title.
- 2 Print the column numbers from 1 to 10.
- 3 Start the outer loop for the rows.
- 4 Print the current row number.
- 5 Start the inner loop for the columns.
- 6 Calculate $i * j$.
- 7 Print the multiplication result.
- 8 Repeat until all 10 columns are printed.
- 9 Move to the next row.
- 10 Repeat until all 10 rows are printed.

💡 Example

◆ For row `3`, the inner loop calculates:

$$3 \times 1 = 3$$

$$3 \times 2 = 6$$

$$3 \times 3 = 9$$

...

$$3 \times 10 = 30$$

➡ Result

The program produces a table similar to:

	1	2	3	4	5	6	7	8	9	10
1	1	2	3	4	5	6	7	8	9	10
2	2	4	6	8	10	12	14	16	18	20
3	3	6	9	12	15	18	21	24	27	30
...										
10	10	20	30	40	50	60	70	80	90	100

Note

- ◆ This program uses nested loops because there are two dimensions:
 - The outer loop controls the rows.
 - The inner loop controls the columns.
- ◆ ``i * j`` produces the multiplication result at the intersection of a row and a column.

⊘ Warning

◆ The program is specifically designed for a table from 1 to 10. To generate a different range, the loop limits would need to be changed.



Comparison

▨ Outer `for` Loop

- ✓ Controls the current row.
- ✓ Runs from 1 to 10.
- ✓ Determines the first number in each multiplication.

▤ Inner `for` Loop

- ✓ Controls the current column.
- ✓ Runs from 1 to 10 for every row.
- ✓ Determines the second number in each multiplication.

✦✦ Summary

- ✓ Nested `for` loops can be used to create a two-dimensional table.
- ✓ The outer loop controls the rows.
- ✓ The inner loop controls the columns.
- ✓ `i \* j` calculates each multiplication result.
- ✓ The same nested-loop pattern can be used for many grid and table problems.

Thank you 😊